



Computational Modeling of Estrogen-Mediated Radioresistance in Three-Dimensional MCF-7 Breast Cancer Spheroids Using PhysiCell

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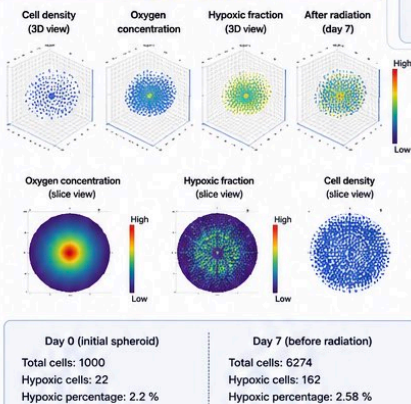
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INTRODUCTION

Breast cancer remains the most frequently diagnosed malignancy among women worldwide. Approximately 70% of breast cancers express the estrogen receptor (ER), making estrogen signaling an important regulator of tumor growth and treatment response. Clinical studies have demonstrated that estrogen enhances cellular proliferation, promotes DNA repair, and reduces apoptosis, thereby contributing to resistance to radiotherapy.

Three-dimensional (3D) tumor spheroids reproduce the structural organization, oxygen gradients, and cell-to-cell interactions observed in solid tumors more accurately than conventional two-dimensional cultures. Computational modeling provides an efficient platform to investigate tumor biology and optimize therapeutic strategies while reducing experimental costs and time.

SIMULATION OVERVIEW



MODEL PARAMETERS (KEY VALUES)

Parameter	Description	Value
Growth rate (λ)	Cell proliferation rate	1.3 day ⁻¹
Linear-Quadratic model	α / β	0.30 / 0.03 Gy ⁻²
Oxygen Enhancement Ratio (OER)	Hypoxia effect on radiation	2.5
Estrogen proliferation factor	Multiplier on proliferation	1.5
Simulation duration	Total simulated time	14 days
Radiation dose (single)	Delivered on Day 7	2 Gy
Time step	Simulation time step	0.01 day

LIMITATIONS

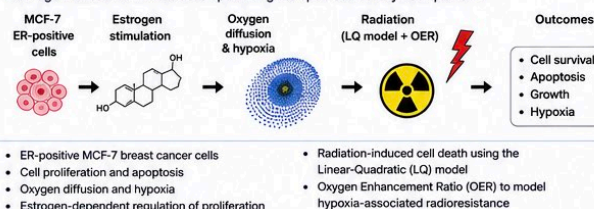
- Simplified representation of the tumor microenvironment.
- Parameters derived from literature may not capture patient variability.
- Does not include immune cells, ECM remodeling, angiogenesis, or stromal cells.
- Intracellular ER signaling represented with simplified assumptions.
- Predictions require experimental validation using 3D MCF-7 spheroids.

USEFUL LINKS

- <https://physicell.org/>
- <https://github.com/MathCancer/PhysiCell>

METHODS

A 3D agent-based model was developed using the open-source PhysiCell platform.



Four experimental conditions were simulated:



RESULTS

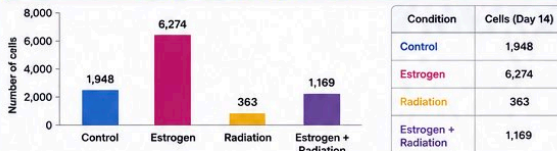
Simulation results demonstrated that estrogen stimulation accelerated spheroid growth and increased viable cell numbers compared with untreated controls. Larger spheroids developed extensive hypoxic cores because oxygen diffusion became limited within the tumor interior.

Following radiation treatment, estrogen-treated spheroids exhibited:

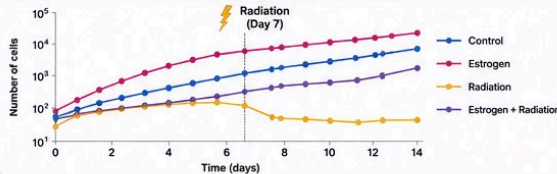
- increased surviving fraction
- reduced apoptosis
- faster post-treatment regrowth
- decreased radiation sensitivity

These findings suggest that estrogen contributes to radioresistance through enhanced proliferation and hypoxia-mediated protection.

VIABLE CELL NUMBER AFTER 14 DAYS



TUMOR GROWTH OVER TIME (VIABLE CELLS)



FUTURE RESEARCH

- Validate the model using in vitro 3D breast cancer spheroids.
- Refine parameters with experimental data.
- Include detailed ER signaling pathways.
- Model endocrine therapies (tamoxifen, fulvestrant).
- Incorporate immune and stromal components.
- Develop patient-specific parameterization for personalized radiotherapy.

CONCLUSION

This study demonstrates the feasibility of using PhysiCell to model estrogen-mediated radioresistance in three-dimensional breast cancer spheroids. The simulation suggests that estrogen promotes tumor growth and decreases radiotherapy effectiveness by enhancing proliferation and increasing hypoxic regions.

Computational modeling represents a valuable tool for investigating cancer biology and designing improved treatment strategies prior to experimental and clinical studies.

SCAN ME

View code & simulations on GitHub



REFERENCES (Selected)

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